

Response to Editorial Feedback

Manuscript: Mechanistic Interpretability of Discrete Logarithm Grokking:
How Representation Determines the Internal Algorithm

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Response to Editorial Feedback

We thank the editorial team for the careful review. Below we address each point in order. All changes have been incorporated into the revised Word manuscript (MLJ_dlp crt_grokking.docx) and the LaTeX/PDF version.

Point 1. Please provide City in the affiliation for author(s) (Aff1) in the manuscript.

Response. The affiliation has been updated to include the city:

Industrial Research at Vesterlund Ventures Holding AB, Stockholm, Sweden

This appears on the title page of both the Word manuscript and the LaTeX/PDF version.

Point 2. Please provide Country affiliation for the author(s) (Aff1) in the manuscript.

Response. The country (Sweden) has been added to the affiliation, as shown in the response to Point 1 above.

Point 3. Discrepancy identified between the submission system and the manuscript for Primary Abstract (~42.07%). Please ensure the Primary Abstract in the submission system and in the manuscript is identical.

Response. The abstract has been rewritten to be shorter and more concise. It is now written in plain text without special characters, making it easy to copy directly into the submission system field. The abstract in the Word manuscript and the abstract in the submission system are now identical. The new abstract focuses on the three main discoveries:

1. Different representations produce different internal algorithms (65% vs. 1% interaction energy).
2. The interaction structure forms during the memorization plateau, well before generalization.
3. Algebraic alignment, not merely factorization, is required for rapid generalization.

Point 4. The location of Table Citation (Table 2) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response. Table 2 (Algebraic Structure Scores) is cited in Section 2.4 (Algebraic Structure Scores). The in-text reference now reads: “The structure scores for all representations are shown in Table 2.” A bold placement marker “[Table 2 about here]” has been added immediately before the table in the Word manuscript.

Point 5. The location of Figure Citation (Fig. 3) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response. Figure 3 (2D CRT Fourier Spectra) is cited in Section 4.2 (Final-State Results). The in-text reference now reads: “The 2D Fourier spectra are visualized in Figure 3, and the

final-state energy values are summarized in Table 5.” A bold placement marker “[Figure 3 about here]” has been added immediately before the figure in the Word manuscript.

Point 6. The location of Table Citation (Table 5) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response. Table 5 (Final-State Fourier Energy) is cited in Section 4.2 (Final-State Results). The in-text reference now reads: “The 2D Fourier spectra are visualized in Figure 3, and the final-state energy values are summarized in Table 5.” A bold placement marker “[Table 5 about here]” has been added immediately before the table in the Word manuscript.

Point 7. The location of Figure Citation (Fig. 2) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response. Figure 2 (Temporal Formation of Interaction Energy) is cited in Section 5.1 (Interaction Energy Forms Before Grokking). The in-text reference now reads: “The temporal trajectory is shown in Figure 2, with the full numerical data in Table 4.” A bold placement marker “[Figure 2 about here]” has been added immediately before the figure in the Word manuscript.

Point 8. The location of Table Citation (Table 4) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response. Table 4 (Temporal Evolution of E_{AB} and Test Accuracy) is cited in Section 5.1 (Interaction Energy Forms Before Grokking). The in-text reference now reads: “The temporal trajectory is shown in Figure 2, with the full numerical data in Table 4.” A bold placement marker “[Table 4 about here]” has been added immediately before the table in the Word manuscript.

Point 9. The location of Figure Citation (Fig. 4) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response. Figure 4 (Top-k Causal Ablation Dose-Response) is cited in Section 6.2 (Results). The in-text reference now reads: “The dose-response curves are shown in Figure 4, and the full ablation results are in Table 3.” A bold placement marker “[Figure 4 about here]” has been added immediately before the figure in the Word manuscript.

Point 10. The location of Table Citation (Table 3) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response. Table 3 (Top-k Causal Ablation Results) is cited in Section 6.2 (Results). The in-text reference now reads: “The dose-response curves are shown in Figure 4, and the full ablation results are in Table 3.” A bold placement marker “[Table 3 about here]” has been added immediately before the table in the Word manuscript.

Point 11. The location of Table Citation (Table 1) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response. Table 1 (Grokking Results Across All Representations) is cited in Section 7.3 (Results). The in-text reference now reads: “The grokking results across all representations are summarized in Table 1. Learning curves for the three key representations (RAW, CRT, R3) are shown in Figure 1.” A bold placement marker “[Table 1 about here]” has been added immediately before the table in the Word manuscript.

Point 12. The location of Figure Citation (Fig. 1) is not clear in the manuscript. Ensure the figure is cited and indicate in the manuscript where it should appear.

Response. Figure 1 (Learning Curves) is cited in Section 7.3 (Results). The in-text reference now reads: “The grokking results across all representations are summarized in Table

1. Learning curves for the three key representations (RAW, CRT, R3) are shown in Figure 1.” A bold placement marker “[Figure 1 about here]” has been added immediately before the figure in the Word manuscript.

Point 13. The location of Table Citation (Table 6) is not clear in the manuscript file. Indicate in the manuscript where the table should appear.

Response. Table 6 (Comparison with the Companion JMLR Manuscript) is cited in Section 9 (Detailed Comparison with the Companion Manuscript). The in-text reference now reads: “The comparison is summarized in Table 6.” A bold placement marker “[Table 6 about here]” has been added immediately before the table in the Word manuscript.

Summary of Changes

All changes have been incorporated into the revised Word manuscript (MLJ_dlp_crt_grokking.docx) and the LaTeX/PDF version (MLJ_dlp_crt_grokking.tex / .pdf).

#	Feedback	Location manuscript	in	Action taken
1	City in affiliation	Title page		Added “Stockholm”
2	Country in affiliation	Title page		Added “Sweden”
3	Abstract discrepancy	Abstract section		Rewritten, shorter, plain text, identical to submission system
4	Table 2 location	Section 2.4		Added in-text citation and “[Table 2 about here]” marker
5	Figure 3 location	Section 4.2		Added in-text citation and “[Figure 3 about here]” marker
6	Table 5 location	Section 4.2		Added in-text citation and “[Table 5 about here]” marker
7	Figure 2 location	Section 5.1		Added in-text citation and “[Figure 2 about here]” marker
8	Table 4 location	Section 5.1		Added in-text citation and “[Table 4 about here]” marker
9	Figure 4 location	Section 6.2		Added in-text citation and “[Figure 4 about here]” marker
10	Table 3 location	Section 6.2		Added in-text citation and “[Table 3 about here]” marker
11	Table 1 location	Section 7.3		Added in-text citation and “[Table 1 about here]” marker
12	Figure 1 location	Section 7.3		Added in-text citation and “[Figure 1 about here]” marker
13	Table 6 location	Section 9		Added in-text citation and “[Table 6 about here]” marker